

UNI Hustle Rwanda: Local Gig Marketplace

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African Leadership University | BSc Software Engineering | Foundations Project

1. Problem, Context, and Users

In Rwanda, 61% of graduates struggle to stand out on crowded global freelance platforms like Upwork, leaving students without a centralized, transparent system to find flexible work. This reliance on informal networks creates massive friction in the local economy: it severely limits earning and portfolio opportunities for ALU students, while simultaneously slowing down the hiring process for local SMEs seeking affordable, high-quality talent.

2. Objectives and Scope

Our main goal is to build a scalable, dual-interface web platform that connects university students directly with local employers.

- Secure Access:** Implement passwordless OTP authentication strictly restricted to verified student emails.
- Seamless Operations:** Automate the backend processing for gig posting, tracking, and application management.
- Actionable Analytics:** Provide a real-time analytical dashboard to help businesses instantly assess their ROI.
- Project Scope:** The initial Beta launch focuses strictly on ALU students and Kigali SMEs, intentionally bypassing large corporate recruitment systems.

3. Project Metadata

Supervisor	Mr. David
Live Link	unihustle-rwanda-core-1186.vercel.app
GitHub Repo	github.com/menes-ux/Unihustle-rwanda-core

The Team

- Ménès Adisso – Project Manager
- Jean Nepo Munezero – Backend Lead
- Gilbert Ntivunwa – Business Analyst
- David Achibiri – Frontend Lead
- Bonheur Munezero – UI/UX Designer
- Manuelle Ackun – Database Architect



Code Repo

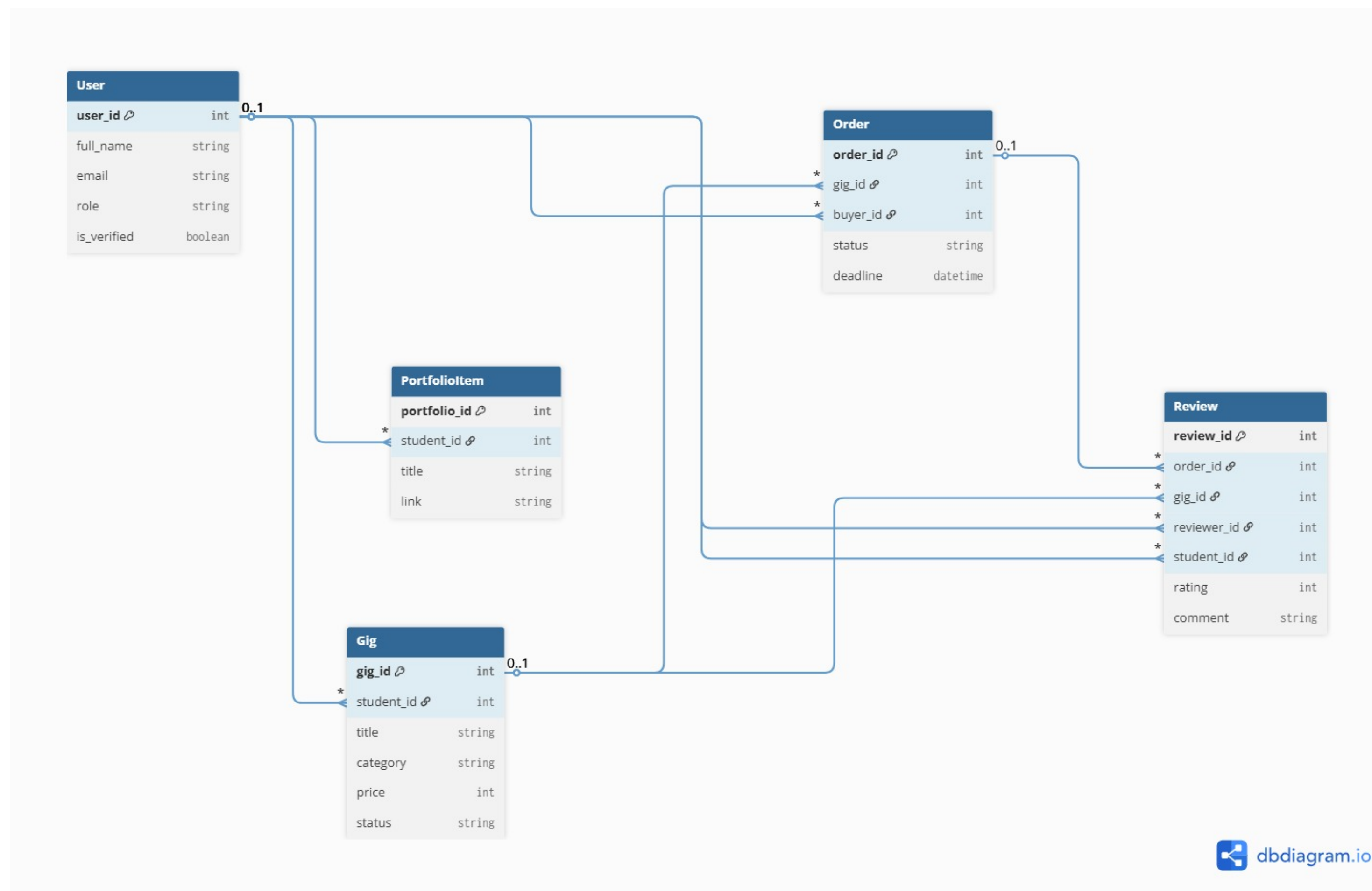
4. Constraints and Design Decisions

To ensure the platform is reliable and secure, we made several strict technical choices:

- Strict Authentication:** We enforced a passwordless OTP login restricted only to `@alustudent.com` emails. This prevents unauthorized access and builds immediate trust with local employers.
- Unified Architecture:** We selected the Next.js App Router. It allows us to handle both the interactive React frontend and the backend API routes in a single, easily deployable codebase.
- Robust Database:** We used Prisma ORM with a PostgreSQL database hosted on Supabase. This guarantees type-safe database queries, meaning fewer runtime errors and better scaling as we add more users.

5. Proposed Solution and System Overview

A secure, responsive two-way marketplace driven by real-time data.

Tech Stack: Next.js (React), Tailwind CSS, Node.js, Prisma ORM, PostgreSQL (Supabase), Vercel.

6. Methodology and Build Process

We adopted a user-centric Agile framework to ensure rapid iteration and strict alignment with both student and employer needs.

Research & System Design

- Conducted targeted interviews with 3 Kigali SMEs and 6 ALU students to define core market friction points.
- Developed high-fidelity, dual-role prototypes in Figma, iterating extensively based on initial user feedback.
- Architected a relational database schema optimized for fast query execution and data integrity.

Engineering & Quality Assurance

- Engineered custom native SVG and Recharts analytics to optimize dashboard rendering and minimize latency.
- Conducted rigorous boundary and security testing on the Nodemailer OTP authentication flow.
- Utilized automated Prisma database seeding to simulate a realistic Beta environment for load and UI testing.

Phase	Implementation Focus
Sprint 1	Discovery, UI/UX Prototyping, and Prisma Schema Architecture
Sprint 2	Next.js App Router Initialization and PostgreSQL Integration
Sprint 3	Core API Logic (Passwordless Auth, Gig Routing, Order Processing)
Sprint 4	Custom Data Visualization, End-to-End Testing, and CI/CD Deployment on Vercel

7. Results and Demo Evidence

70+

Registered Users

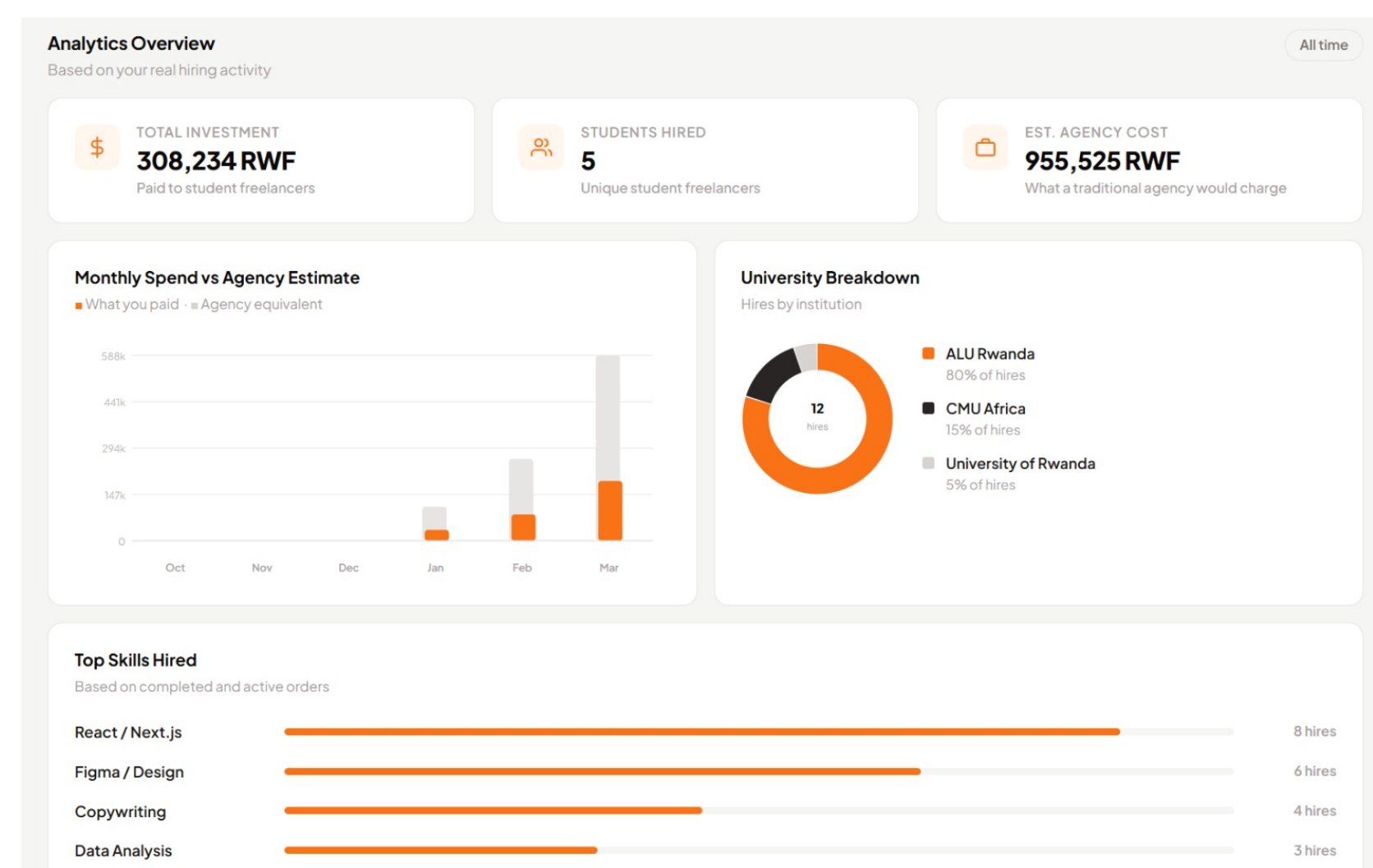
95%

OTP Success Rate

100%

System Uptime

- User Growth:** Steady increase in registrations during the curated Beta testing window.
- System Stability:** OTP verification processes achieved a near-perfect success rate.
- Business ROI:** Dashboard accurately monitors active hires, investment, and agency savings.
- Data Visualization:** Custom native SVGs and Recharts flawlessly render live metrics.



8. Impact and Contribution

- Technical:** Successfully built a full-stack, dual-role platform using modern web technologies.
- User Impact:** Makes it safe and easy for students to earn money without facing global competition.
- Mission:** Contributes directly to Rwanda's Vision 2050 goal of creating 300,000 digital jobs.

Empowering local talent through transparent technology.

9. Limitations and Next Steps

- Current Limitation:** The platform currently lacks in-app messaging between users.
- Next Step 1:** Develop a mobile application (80% of students access the web via phones).
- Next Step 2:** Integrate local payment gateways (e.g., MoMo, Stripe) for seamless transactions.
- Next Step 3:** Build an AI-powered algorithm for predictive student-skill matching.

10. References & Acknowledgements

- World Bank (2022). *Rwanda economic update: Digital transformation for inclusive growth*.
- ILO (2021). *The role of digital labour platforms in transforming the world of work*.
- Niyonsenga, T. (2022). Challenges of freelance work among Rwandan graduates. *Rwanda Journal of Social Sciences*, 9(2).

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